

ABITATHA ROY

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Headline

Undergraduate Researcher in Data Science and Machine Learning, with experience in Cross-Domain Generalization, Healthcare Prediction, and Dynamical Systems. Interested in Large Language Models and Multimodal Learning.

Education

B.Tech. Information Technology and Mathematical Innovations

Expected 2027

Cluster Innovation Centre, University of Delhi

Score: 89.8%

Selected Coursework: Calculus, Linear Algebra, ODE, Linear Programming and Game Theory, Probability and Statistics, DBMS, Data Science, Nanotechnology, *in silico* Biology, Genes to Genomes

Indian School Certificate (ISC) (Class XII)

2023

Calcutta Public School, Kalikapur, Kolkata

Score: 97.0%

Selected Experience

Cluster Innovation Centre, University of Delhi

New Delhi, India

Student Intern – Supervisor: Prof. Shobha Bagai

09/2025 – 11/2025

- Title: Analysis of Air Quality Index Prediction and Pollution Dynamics Across Indian Cities with a Focus on Delhi. Publication under preparation.
- Conducted studies across 4 Indian cities using a single representative weather station, followed by studies across 13 stations across Delhi.
- Analyzed ML algorithms like RF, XGBoost and CatBoost for AQI prediction and the variance of top contributing pollutants (like PM2.5, PM10) and meteorological factors by region.
- Analyzed how wind patterns contribute towards pollutant accumulation and dispersion.

Achievements

2nd Position – MathWorks CodeMatrix Sprint

NIT Delhi, India

Tech Stack: MATLAB, Statistics and Machine Learning Toolbox, Symbolic Math Toolbox

10/2025

- Built a Gamified Fitness Tracker approach linking with Indian Railways storyline.
- Captures and processes real-time mobile sensor data and integrates machine learning to calculate and report on user kinematics (e.g. speed, distance) and calories burned, and integrates gamification module.

Selected Projects

Cross-Dataset Performance Evaluation of Predictive Algorithms for Phishing Website Classification

Tech Stack: Python, Scikit-learn, XGBoost, Pandas

Supervisor: Prof. Shobha Bagai

- Implemented standardized URL feature extraction pipeline (17 features); identified optimal 9-feature subset via mutual information, achieving 84.55% cross-dataset accuracy with Random Forest.
- Conducted cross-dataset evaluation (Mendeley → PhiUSIIL) to study domain shift; implemented dataset harmonization to align feature distributions.

Predicting Tuberculosis Treatment Default Using Artificial Neural Networks

Tech Stack: Python, TensorFlow/Keras, Streamlit, Pandas

Supervisor: Dr. Harendra Pal Singh

- Integrated three TB datasets; addressed domain variability via feature harmonization.
- Designed regularized ANN (dropout + L2) optimized for high recall in default prediction.
- Achieved ROC-AUC ≈ 0.75 and PR-AUC ≈ 0.89 , prioritizing sensitivity to defaults.
- Deployed interactive Streamlit dashboard for real-time clinical risk assessment. Publication under preparation.

Technical report and reproducible code available upon request.

Selected Certifications

IBM Data Science Professional Certificate – Coursera

Bioinformatics Specialization by UC San Diego – Coursera (Ongoing)

Skills

Core Areas: Probabilistic Modelling, Data Science, Machine Learning, Artificial Neural Networks, Bioinformatics, Large Language Models, Prompt Engineering, Dynamical Systems, Applied Mathematics

Programming and Tools: Python, R, C, Java, MySQL, Scikit-learn, TensorFlow/Keras, Streamlit, Flask, MATLAB, Mathematica, LTspice, Pandas, NumPy, MS Excel